tRNA introns as mRNA regulators: A function for a forgotten molecule

Introduction

Transfer RNAs (tRNAs) are small noncoding RNAs that are essential for translation in all kingdoms of life[1]. Defects in tRNA biology cause various human disorders ranging from neurodegenerative diseases to cancer[1]. In eukaryotes, a tRNA begins its journey after transcription from tRNA genes. The tRNA then undergoes extensive processing, modification, and intracellular transit prior to acting in translation[1]. Some tRNA genes contain an intron, which necessitates an additional tRNA processing step: intron removal (splicing). Splicing generates a mature tRNA and a free tRNA intron. Because free tRNA introns are rapidly degraded[1], and deletion of tRNA introns causes minimal growth defects in yeast[2], it was assumed that tRNA introns have no biological function.

However, Dr. Gina Nostramo (a postdoctoral fellow in my lab) and I, along with other lab colleagues, discovered that free tRNA introns act as a novel class of noncoding regulatory RNAs that target complementary mRNA coding regions (Figure 1), revealing a previously unknown form of gene regulation and opening a new realm of tRNA biology. A paper detailing our findings[3], on which I am second author, is in press at the journal *Molecular Cell*. Described below are the experiments leading up to this discovery and my next directions in the nascent tRNA intron field.

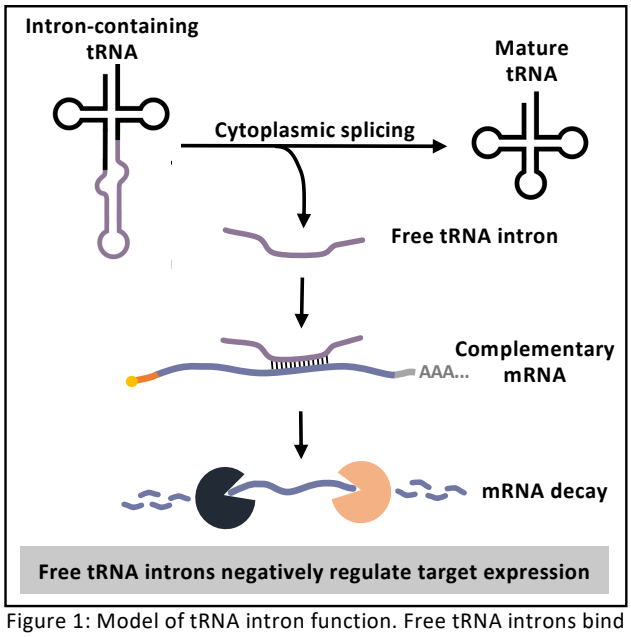

Figure 1: Model of tRNA intron function. Free tRNA introns bind complementary mRNA coding regions, causing mRNA decay.

Discovery of tRNA intron function

When I joined Dr. Anita Hopper's lab, prior data in the lab suggested that tRNA introns may have a cellular function. First, in the budding yeast *S. cerevisiae,* tRNA intron turnover is stress-responsive and occurs through at least five distinct pathways. Second, the presence of tRNA introns is conserved. Every eukaryotic organism has at least one intron-containing tRNA family[4].

During a lab meeting two years ago, Dr. Nostramo presented preliminary data on a third facet of tRNA introns: their complementarity to mRNAs and their apparent regulation of those mRNAs. Regarding this potential function for tRNA introns, I had various questions: Is the complementarity between tRNA introns and mRNAs occurring by chance? Does it extend beyond the tRNA introns to the tRNA exons? Do tRNA introns in other organisms show similar attributes? Encouraged by Drs. Hopper and Nostramo to pursue these questions, I began an exciting scientific journey.

Initially, I used existing bioinformatics tools to investigate tRNA intron complementarity, but found they lacked the functionality needed to fully address my questions. Motivated to overcome this limitation, I taught myself to code in Python and developed custom programs to scan the yeast genome and identify regions of mRNA complementarity to tRNA introns. While I investigated all 10 tRNA intron families, we chose to focus our studies on the 60 nucleotide $tRNA^{Ile}$ intron, which is the longest in yeast. Through my analyses, I learned that the $tRNA^{Ile}$ intron has long stretches of

complementarity (≥13/13 or ≥14/15 nucleotides) to the coding regions of a remarkable 33 mRNAs. Diving deeper into intron complementarity, I uncovered several findings: **(1)** I demonstrated that the degree of complementarity exhibited by the $tRNA^{Ile}$ intron (as well as other tRNA introns) is exceptional. Out of 1000 computer-generated sequences at the same length and GC content as the $tRNA^{Ile}$ intron, <2% have complementarity to as many mRNAs as the $tRNA^{Ile}$ intron. **(2)** By launching our lab onto the Ohio Supercomputer, I was able to efficiently run code to scramble the yeast genome while preserving codon usage. This revealed that the number of $tRNA^{Ile}$ intron targets was reduced in 98% of scrambled genomes, indicating that tRNA intron complementarity is dependent on the natural codon arrangement of the genome. **(3)** I found that mRNA complementarity is largely confined to tRNA introns, with mature tRNAs having minimal complementarity despite their larger size (Figure 2). Together, the computational data highlight that tRNA introns display remarkable and selective complementarity to mRNA coding regions.

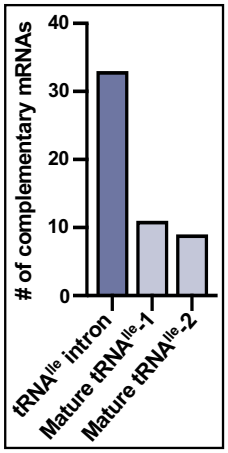

Figure 2: Number of mRNA coding regions with ≥13/13 or ≥14/15-nts of complementarity to the $tRNA^{Ile}$ intron (60nts) and the mature $tRNA^{Ile}$ (73nts).

These findings led to the hypothesis that tRNA introns in *S. cerevisiae* may interact with complementary mRNA sequences similarly to regulatory microRNAs that are found in other organisms (although different since microRNAs target 3' UTR sequences[5]). To test the hypothesis, Dr. Nostramo generated a strain of yeast that lacks the $tRNA^{Ile}$ introns from both $tRNA^{Ile}$ genes (*IleiΔ*). In the *IleiΔ* strain, there are significant increases in the levels of the mRNAs with complementarity to the $tRNA^{Ile}$ intron relative to mRNAs lacking complementarity. This supports an inhibitory role for tRNA introns on complementary mRNAs. Congruently, inducible overexpression of the $tRNA^{Ile}$ intron results in a reduction of mRNAs with complementarity to the intron relative to mRNAs lacking complementarity. Mutations in the tRNA intron that disrupt regions of mRNA complementarity nullify the effect on the respective mRNA, supporting a direct interaction. While these findings indicate that tRNA introns are direct inhibitors of complementary mRNAs levels, the boundaries of the effect remained unclear.

To investigate the scope of regulation by tRNA introns, I took three approaches. First, I used published transcript isoform sequencing data[6], to identify a testable set of mRNAs with $tRNA^{Ile}$ intron complementarity in their 3' UTRs. RT-qPCR revealed that the levels of these mRNAs are unaffected upon intron deletion or overexpression. This differentiates regulation by tRNA introns from microRNAs, which canonically target the 3' UTRs of mRNAs[5]. Interestingly, this is consistent with *S. cerevisiae* lacking the argonaute proteins necessary for microRNA function[7], indicating that mRNA regulation by tRNA introns occurs through an alternative, currently unknown, mechanism. Second, to investigate whether tRNA introns can regulate mRNAs while still housed in the pre-tRNA, I identified mRNA coding regions with complementarity overlapping tRNA intron-exon junctions. I ran RT-qPCR to measure the levels of these mRNAs in a *tom70Δ* strain (which is defective in tRNA splicing and accumulates intron-containing pre-tRNAs[8]), finding no change relative to mRNAs lacking complementarity. Jointly with other evidence, this demonstrates that tRNA introns do not regulate complementary mRNAs while still housed in the pre-tRNA. Third, I used NCBI databases to investigate tRNA introns across organisms, uncovering that tRNA introns, and many of their target regions, are highly conserved in related yeast species. Expanding tRNA intron analysis

to humans, I utilized the Ohio Supercomputer to find that human tRNA introns have long stretches of complementarity (≥13/13 nucleotides) to over 100 human mRNAs, raising the possibility that the regulatory role of tRNA introns extends to higher eukaryotes.

Collectively, our findings establish tRNA introns as the newest class of regulatory RNAs, but the rules governing tRNA intron function and the physiological importance of tRNA introns are still unknown. Therefore, I aim to **1.** *Investigate the complementarity required for tRNA intron function* and **2.** *Explore the role of tRNA introns in cellular stress responses.* Over the summer, I was an intern in Dr. Sandra Wolin's lab at the NIH, where I learned how to perform endogenous genetic modifications. I will now apply this experience alongside the skills I continue to refine in Dr. Hopper's lab, such as handling yeast, isolating RNA, Northern blot, and RT-qPCR, to achieve these aims.

Aim 1: MicroRNAs require less sequence complementarity than the 13-15 nucleotide parameters studied in tRNA introns, possessing regulatory functionality with as few as seven complementary nucleotides[5].This raises the possibility that tRNA introns might also function with less complementarity than we have tested. To explore this, I plan to (1) modify regions of tRNA intron complementarity on mRNAs to examine how changes in nucleotide complementarity alter regulatory effects, and (2) perform RNA sequencing to assess the transcriptome-wide impact of tRNA intron deletion and overexpression. This analysis will focus on mRNAs with short regions of tRNA intron complementarity, which my previously developed programs can efficiently identify.

Aim 2: Under oxidative stress, we observe 15-fold accumulation of the $tRNA^{Trp}$ intron, resulting in decreased levels of complementary mRNAs[3]. This suggests that the $tRNA^{Trp}$ intron plays a role in the response to oxidative stress. To investigate if tRNA introns, like other classes of regulatory RNAs[9], are broader modulators of cellular stress responses, I will use Northern blot to measure the effects of a diverse set of stresses on tRNA intron levels. This will be performed in collaboration with a technician in the lab. For each stress-responsive tRNA intron, I will use yeast stress-response databases, tRNA intron complementarity data, and gene ontology tools to identify potential downstream molecular roles. If successful, these experiments will offer further insights into the connection between cellular stress and tRNA introns, both advancing our fundamental understanding of cellular stress and potentially opening a new avenue to study the many diseases caused by dysregulated cellular stress responses.

These two aims will guide my final undergraduate year, after which I will pursue a Ph.D. to further probe the world of RNA and take the next steps on my research journey.

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